

PL Assignment II

August 26, 2025

Q1. Given any natural number n , show that there is a bijection between the set of all truth functions from $\{T, F\}^n$ to $\{T, F\}$ and the quotient set $\mathcal{F}_n / \equiv_n$.

Q2. Prove the following for any set $\Gamma \cup \Delta \cup \{\alpha\}$ of wffs in PC.

1. $\Gamma \models \alpha$, if and only if $\Gamma \cup \{\neg\alpha\}$ is contradictory.
2. If Γ is satisfiable and $\Delta \subseteq \Gamma$ then Δ is satisfiable.
3. If Γ is contradictory and $\Gamma \subseteq \Delta$ then Δ is contradictory.
4. (Overlap) If $\alpha \in \Gamma$ then $\Gamma \models \alpha$.
5. (Dilution) If $\Gamma \models \alpha$, $\Gamma \subseteq \Delta$ then $\Delta \models \alpha$.
6. (Cut) If $\Gamma \models \alpha$ and $\Delta \models \beta$, for each $\beta \in \Gamma$ then $\Delta \models \alpha$.
7. $\Gamma \cup \{\alpha\} \models \beta$, if and only if $\Gamma \models \alpha \rightarrow \beta$.

Q3. A set Γ of wffs is said to be *finitely satisfiable*, if all its finite subsets are satisfiable. Prove that the following are equivalent.

1. Γ is satisfiable, if and only if Γ is finitely satisfiable.
2. Γ is contradictory, if and only if there is a finite subset of Γ that is contradictory.
3. For any wff α , $\Gamma \models \alpha$, if and only if there is a finite subset Δ of Γ such that $\Delta \models \alpha$.

Q4. Show that $\{\neg, \wedge\}$, $\{\neg, \vee\}$ and $\{\neg, \rightarrow\}$ are all adequate sets of connectives.

Q5. Show that none of the following is an adequate set of connectives.

1. $\{\wedge, \vee, \rightarrow\}$;

2. set containing any one of the connectives $\wedge, \vee, \rightarrow$;
3. set containing any two of the connectives $\wedge, \vee, \rightarrow$.

Q6. Let $*$ be a binary connective such that $\{*\}$ is adequate. Prove that $f_*(T, T) = F$ and $f_*(F, F) = T$.

Q7. Show that the *only* adequate binary connectives are *NOR* and *NAND*.

Q8.(*Negation normal form*) The set of wffs in *negation normal form* (NNF) is defined as follows:

[F1] p and $\neg p$ are in NNF for every $p \in PV$;

[F2] If α and β are in NNF, so are $\alpha \wedge \beta$ and $\alpha \vee \beta$.

[F3] Every wff in NNF is obtained by a finite number of applications of **F1** and **F2**.

The only connectives that occur in wffs in NNF are $\neg, \wedge$ and $\vee$; moreover the only occurrences of $\neg$ are before propositional variables.

Note that $p \vee \neg q$ is in NNF but $\neg(p \vee \neg q)$ is not. But $\neg(p \vee \neg q)$ is logically equivalent to $\neg p \wedge q$, which is in NNF.

Show that every wff is logically equivalent to a wff in NNF.